

Basil Peter

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Location: India

PROFESSIONAL SUMMARY

Computer Science Engineering graduate (2025) and Full Stack Developer with a strong focus on **system design, adaptive logic, and explainable intelligence**. Experienced in building full-stack applications that go beyond CRUD by incorporating **behavior modeling, feedback loops, and learning-driven decision systems**. Proficient in React, TypeScript, Python, Flask, and modern AI integrations, with an emphasis on clarity, debuggability, and real-world reasoning.

TECHNICAL SKILLS

Languages	JavaScript (ES6+), TypeScript, Python, SQL, HTML5, CSS3
Frontend	React.js, Vite, Tailwind CSS, Responsive UI Design
Backend	Node.js, Express.js, Flask, RESTful APIs
Databases	MongoDB, PostgreSQL, SQLite, Supabase
AI / Intelligent Systems	Google Gemini API, OCR (Tesseract), Semantic Embeddings, Explainable Decision Logic
Tools	Git, GitHub, VS Code, Postman

PROJECTS

IntentOS – Adaptive Intent-Driven Learning System (System Design + Full Stack) [GitHub](#)

- Designed a closed-loop system that infers user intent from behavioral signals and dynamically adapts application flow
- Implemented an explainable intent inference engine using rule-based temporal signal analysis
- Built a policy-driven adaptation layer to modify UI behavior without hardcoded logic
- Developed an outcome-based learning engine to reinforce or decay policies based on success and drop-off feedback
- Created an explainability dashboard visualizing live signals, intent transitions, policy performance, and learning state

Capstone Project: EchoForm – Cognitive Identity Modeling System (Full Stack + ML) [GitHub](#)

- Designed a probabilistic engine to infer evolving user behavioral traits from textual interaction data
- Implemented semantic signal extraction using sentence embeddings and cosine similarity
- Built belief update pipelines incorporating temporal decay, volatility tracking, and uncertainty estimation
- Introduced active learning through clarifying questions to resolve high-entropy inference states
- Developed dashboards to visualize confidence drift and behavioral volatility over time

Blood Bank Management System (MERN Stack) [Live](#) — [GitHub](#)

- Developed a full-stack platform connecting donors, hospitals, and blood banks
- Implemented JWT-based authentication and role-based access control
- Enabled real-time blood inventory tracking to reduce emergency request response time

SyncVerse AI – AI-Powered Knowledge Platform (Personal Project) [Live](#) — [GitHub](#)

- Built a platform to chat with PDFs, documents, and YouTube transcripts using contextual AI
- Integrated Google Gemini API for multi-document question answering

- Implemented OCR fallback using Tesseract to support scanned PDFs

EventFlow – Event Equipment Rental Platform

[Live](#) — [GitHub](#)

- Built a rental platform with real-time availability and interactive cart functionality
- Developed using React, TypeScript, Supabase, and Tailwind CSS

EDUCATION

Bachelor of Technology (B.Tech) – Computer Science Engineering

2021 – 2025

KMCT College of Engineering